

SHENGYU FENG

Homepage: <https://shengyu-feng.github.io>

Google Scholar $\diamond$ Github $\diamond$ LinkedIn

EDUCATION

Carnegie Mellon University (CMU)

Aug. 2022 – Present

Ph.D. in Language and Information Technology, GPA: 4.06/4.30

Advisor: Yiming Yang

University of Illinois at Urbana-Champaign (UIUC)

Aug. 2020 – May 2022

M.S. in Computer Science, GPA: 3.91/4.00

Advisor: Hanghang Tong

University of Michigan (UM)

Aug. 2018 – May 2020

B.S.E in Computer Science, GPA: 4.00/4.00

Shanghai Jiao Tong University (SJTU)

Sep. 2016 – Aug. 2020

B.S in Electrical and Computer Engineering, GPA: 3.67/4.00

RESEARCH INTERESTS

My research focuses on **learning under limited data, supervision, and feedback**, with a focus on **optimization, generative models and reinforcement learning** algorithms for foundation models.

WORK EXPERIENCE

Apple Foundation Model Team

May 2026 – Aug. 2026

Research Intern (*LLM pretraining, optimization, and scaling*)

Meta Superintelligence Labs

May 2025 – Aug. 2025

Research Intern (*Generative verifier*)

Apple Foundation Model Team

May 2024 – Aug. 2024

Research Intern (*LLM math reasoning*)

Microsoft Research

May 2022 – Aug. 2022

Research Intern

Intel AI Lab

May 2021 – Aug. 2021

Graduate Research Intern

SELECTED PUBLICATIONS (GROUPED BY RESEARCH AREAS)

Foundation Models & Reinforcement Learning

- Beyond Data and Model: Optimization-Aware Scaling under Constrained Data. **Shengyu Feng**, Xiang Kong, Zhifeng Chen, Yiming Yang. *Preprint*.
- Rubric-Based Benchmarking and Reinforcement Learning for Advancing LLM Instruction Following. Yun He*, Wenzhe Li*, Hejia Zhang, Songlin Li, Karishma Mandyam, Sopan Khosla, Yuanhao Xiong, Nanshu Wang, Selina Peng, Beibin Li, Shengjie Bi, Shishir G. Patil, Qi Qi, **Shengyu Feng**, Julian Katz-Samuels, Richard Yuanzhe Pang, Sujun Gonugondla, Hunter Lang, Yue Yu, Yundi Qian, Maryam Fazel-Zarandi, Licheng Yu, Amine Benhalloum, Hany Awadalla, and Manaal Faruqui. *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

- Bradley-Terry Policy Optimization for Generative Preference Modeling. **Shengyu Feng**, Yun He, Shuang Ma, Beibin Li, Yuanhao Xiong, Vincent Li, Karishma Mandyam, Julian Katz-Samuels, Shengjie Bi, Licheng Yu, Hejia Zhang, Karthik Abinav Sankararaman, Han Fang, Yiming Yang, and Manaal Faruqui. *Preprint*.
- Step-by-Step Reasoning for Math Problems via Twisted Sequential Monte Carlo. **Shengyu Feng**, Xiang Kong, Shuang Ma, Aonan Zhang, Dong Yin, Chong Wang, Ruoming Pang, and Yiming Yang. *International Conference on Learning Representations (ICLR)*, 2025.
- SORREL: Suboptimal-Demonstration-Guided Reinforcement Learning for Learning to Branch. **Shengyu Feng** and Yiming Yang. *AAAI Conference on Artificial Intelligence (AAAI)*, 2025 (*Oral, 6% of submissions*).
- Batch Reinforcement Learning Through Continuation Method. Yijie Guo, **Shengyu Feng**, Nicolas Le Roux, Ed Chi, Honglak Lee, and Minmin Chen. *International Conference on Learning Representations (ICLR)*, 2021.
- Memory Based Trajectory-conditioned Policies for Learning from Sparse Rewards. Yijie Guo, Jongwook Choi, Marcin Moczulski, **Shengyu Feng**, Samy Bengio, Mohammad Norouzi, and Honglak Lee. *Neural Information Processing Systems (NeurIPS)*, 2020.

Generative Modeling & Optimization

- Unsupervised Diffusion Solver for Combinatorial Optimization via Combinatorial Adjoint Matching. **Shengyu Feng**, Tarun Suresh, and Yiming Yang. *International Conference on Machine Learning (ICML)*, 2026.
- Unsupervised Neural Langevin Sampler for Mixed Integer Linear Programming. Yixin Huang, **Shengyu Feng**, and Yiming Yang. *International Conference on Machine Learning (ICML)*, 2026.
- Regularized Langevin Dynamics for Combinatorial Optimization. **Shengyu Feng** and Yiming Yang. *International Conference on Machine Learning (ICML)*, 2025.

Evaluation & Agents

- FrontierCO: Real-World and Large-Scale Evaluation of Machine Learning Solvers for Combinatorial Optimization. **Shengyu Feng***, Weiwei Sun*, Shanda Li, Ameet Talwalkar, and Yiming Yang. *International Conference on Learning Representations (ICLR)*, 2026.
- CO-Bench: Benchmarking Language Model Agents in Algorithm Search for Combinatorial Optimization. Weiwei Sun*, **Shengyu Feng***, Shanda Li, and Yiming Yang. *AAAI Conference on Artificial Intelligence (AAAI)*, 2026.

Representation & Self-Supervised Learning

- ARIEL: Adversarial Graph Contrastive Learning. **Shengyu Feng**, Baoyu Jing, Yada Zhu, and Hanghang Tong. *ACM Transactions on Knowledge Discovery from Data (TKDD)*.
- X-GOAL: Multiplex Graph Prototypical Contrastive Learning. Baoyu Jing, **Shengyu Feng**, Yuejia Xiang, Xi Chen, Yu Chen, and Hanghang Tong. *ACM International Conference on Information and Knowledge Management (CIKM)*, 2022.
- Adversarial Graph Contrastive Learning with Information Regularization. **Shengyu Feng**, Baoyu Jing, Yada Zhu, and Hanghang Tong. *ACM Web Conference (WWW)*, 2022.

Distributed Computing

- Alternative Mixed Integer Linear Programming Optimization for Joint Job Scheduling and Data Allocation in Grid Computing. **Shengyu Feng***, Jaehyung Kim*, Yiming Yang, Joseph Boudreau, Tasnuva Chowdhury, Adolfo Hoisie, Raees Khan, Ozgur O. Kilic, Scott Klasky, Tatiana Korchuganova, Paul Nilsson, Verena Ingrid Martinez Outschoorn, David K. Park, Norbert Podhorszki, Yihui Ren, Frédéric Suter, Sairam Sri Vatsavai, Wei Yang, Shinjae Yoo, Tadashi Maeno, and Alexei Klimentov. *Future Generation Computer Systems (FGCS)*.

HONORS & AWARDS

Sansom Graduate Fellowship, CMU	2026
LTI Exceptional Service Award, CMU	2026
OpenAI Researcher Access Program Grant	2025
Siebel Scholars for class 2022, UIUC	2021
University Merit Student, SJTU	2017, 2018
Interdisciplinary Contest in Modeling (ICM) Meritorious Winner	2017

TEACHING EXPERIENCE

Teaching Assistant, CMU 11441/11741: Machine Learning with Graphs Instructor: Yiming Yang	<i>Fall 2024, Fall 2025</i>
Teaching Assistant, UIUC CS 445: Computational Photography Instructor: Derek Hoime	<i>Fall 2020, Spring 2021</i>
Instructional Aide, UM EECS 442: Computer Vision Instructor: David Fouhey (Fall 2019) and Justin Johnson (Winter 2020)	<i>Fall 2019, Winter 2020</i>

PROFESSIONAL SERVICE

Organizer, CMU LLM Agent Workshop	2025
Reviewer, Neural Information Processing Systems (NeurIPS)	2021 – 2025
Reviewer, International Conference on Machine Learning (ICML)	2022 – 2025
Reviewer, International Conference on Learning Representations (ICLR)	2022 – 2024

INVITED TALKS

Regularized Langevin Dynamics for Combinatorial Optimization 2025 INFORMS Annual Meeting, oral presentation	<i>Oct. 2025</i>
Benchmarking LLM Agents in Algorithm Search Massachusetts Institute of Technology (MIT), invited benchmark talk	<i>April 2025</i>

LEADERSHIP & ENGAGEMENT

Graduate Student Assembly Representative, CMU	2023 – 2026
Member of Campus Affair Committee, CMU	2023 – 2025
Member of UM-SJTU Alumni Association, SJTU	2017 – 2020